

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Case Study

COURSE NAME: COMPUTER VISION COURSE CODE:ITA0515 Slot B

COURSE OUTCOME COVERED

CO3: Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BTL 4)

Assessment Tool Weightage: 40%

QUESTIONS:5

Total Marks 50

Duration :60

Minutes

Annexure A

Case Study

Code of Conduct:

I R Dinesh-192421117 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Rubrics for Evaluation

Criteria	Weight age (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks(100)
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	

Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

1. Preprocessing and Feature Detection

A vision system processes images captured under varying lighting conditions. Without preprocessing, feature detection results are inconsistent. Analyze the issue and explain how preprocessing improves feature detection performance.

2. Edge vs Corner Detection

In an image containing both straight edges and textured regions, a system uses edge detection but fails to identify key points for matching. Analyze why edge detection is insufficient and justify the use of corner detection in this case.

3. Hough Transform in Noisy Images

A system uses Hough Transform to detect lines in noisy images but produces inaccurate results.

Analyze the effect of noise on detection and suggest improvements based on preprocessing or parameter selection.

4. Feature Matching under Transformations

Two images of the same object are taken at different scales and orientations. Traditional feature matching fails, but scale invariant feature matching works effectively. Analyze why scale invariant methods perform better in this scenario.

5. PCA for Feature Optimization

A system extracts a large number of features, leading to high computational cost and redundancy.

Analyze how Principal Component Analysis helps in reducing dimensionality and improving efficiency.

1. Preprocessing and Feature Detection

Case Study Title

Improving Face Recognition Accuracy Under Varying Lighting Conditions

1. Problem Statement

A college security system uses a camera-based face recognition system to identify students entering the campus. The system works accurately under normal lighting but produces inconsistent results when images are captured in bright sunlight, low light, or uneven illumination.

2. Scenario

The same student may be captured under different lighting conditions:

- Bright sunlight
- Indoor lighting
- Low-light conditions
- Shadows on the face

These conditions change the intensity and contrast of facial features.

3. Analysis

Feature detection algorithms depend on image intensity variations. When illumination changes, important features such as the eyes, nose, and facial boundaries may appear different.

As a result:

- Some features may not be detected.
- False features may be detected.
- Feature matching becomes less reliable.

4. Proposed Solution

A preprocessing stage is introduced before feature detection:

Input Image → Grayscale Conversion → Noise Removal → Contrast Enhancement → Normalization → Feature Detection

Techniques such as **Gaussian filtering, histogram equalization, and contrast enhancement** can improve the image quality.

5. Expected Result

After preprocessing, the important facial features become more consistent across different lighting conditions. The feature detector can therefore identify more reliable feature points.

6. Conclusion

Image preprocessing is an important stage in a vision system because it reduces the effect of illumination and noise. This improves the consistency and reliability of subsequent feature detection.

2. Edge vs Corner Detection

Case Study Title

Building Recognition Using Distinctive Feature Points

1. Problem Statement

A mobile vision application is designed to recognize buildings from photographs. The images contain walls, windows, doors, and textured architectural patterns. The system initially uses only edge detection but fails to obtain sufficient key points for reliable image matching.

2. Scenario

Edge detection successfully identifies:

- Wall boundaries
- Window boundaries
- Door boundaries
- Roof edges

However, many points along the same straight edge have similar characteristics.

3. Analysis

An **edge** represents a significant intensity change primarily in one direction. Therefore, an edge does not necessarily provide a unique location for matching.

A **corner**, on the other hand, represents significant intensity changes in two directions. Corners are therefore more distinctive and suitable as feature points.

4. Proposed Solution

The system should use **Harris Corner Detection** after preprocessing.

Building Image → Preprocessing → Harris Corner Detection → Key Points → Feature Matching

Corners can be detected at:

- Window intersections
- Door corners
- Wall intersections
- Architectural junctions

5. Expected Result

The system obtains distinctive key points that can be matched between photographs of the same building.

6. Conclusion

Edge detection is suitable for identifying object boundaries, but it is insufficient when unique feature points are required. Harris Corner Detection provides more distinctive points and is therefore more appropriate for image matching.

3. Hough Transform in Noisy Images

Case Study Title

Improving Road Lane Detection in Noisy Camera Images

1. Problem Statement

An autonomous vehicle uses a camera to identify road lanes using the Hough Transform. Under clear conditions, lane detection works well. However, rain, shadows, road texture, and camera noise cause inaccurate line detection.

2. Scenario

The input image may contain:

- Road surface texture
- Shadows
- Rain
- Cracks
- Vehicle boundaries
- Sensor noise

These unwanted structures can be interpreted as edges.

3. Analysis

The Hough Transform detects lines based on votes from edge pixels. When noise produces additional edges, these pixels also contribute votes.

This can result in:

- False lines
- Incorrect lane positions
- Multiple unwanted lines
- Missing lane segments

4. Proposed Solution

A preprocessing and parameter-optimization stage is introduced:

Input Image → Gaussian Filtering → Canny Edge Detection → Hough Transform → Lane Detection

The following parameters can also be adjusted:

- **Threshold** – controls the minimum votes required.
- **minLineLength** – removes very short lines.
- **maxLineGap** – connects broken line segments.
- **rho and theta** – control spatial and angular resolution.

5. Expected Result

Noise is reduced before Hough processing, while appropriate parameters eliminate weak and irrelevant line detections. The actual lane markings can therefore be detected more accurately.

6. Conclusion

The accuracy of Hough Transform depends strongly on the quality of the edge image and parameter selection. Preprocessing combined with proper parameter tuning improves line detection in noisy environments.

4. Feature Matching Under Transformations

Case Study Title

Product Recognition Under Scale and Rotation Changes

1. Problem Statement

An e-commerce application allows users to photograph a product and search for the same product in an online database. Traditional feature matching fails when the photograph is taken from a different distance or orientation.

2. Scenario

The database image may show a product:

- At normal scale
- In a horizontal orientation

The user's photograph may show the same product:

- Closer or farther from the camera
- Rotated
- At a different orientation

3. Analysis

Traditional feature descriptors may depend on the original size and orientation of the detected features. Changes in scale and rotation can therefore make corresponding features difficult to match.

Scale-invariant methods are specifically designed to produce features that remain recognizable under such transformations.

4. Proposed Solution

Use a **scale- and rotation-invariant feature detection and description method**, such as SIFT.

Database Image → Feature Extraction

User Image → Feature Extraction → Scale/Rotation-Invariant Descriptors → Feature Matching → Product Identification

5. Expected Result

Even when the product appears at a different scale or orientation, corresponding feature points can still be identified and matched.

6. Conclusion

Scale-invariant feature matching is more robust than traditional matching because it handles changes in object size and orientation. It is therefore suitable for product recognition, image retrieval, and object identification.

5. PCA for Feature Optimization

Case Study Title

Reducing Feature Dimensionality in a Face Recognition System

1. Problem Statement

A face recognition system extracts a very large number of features from every face image. Processing all these features increases computational cost and introduces redundancy.

2. Scenario

Suppose the system extracts:

1,000 features per face image

Many of these features may contain similar or correlated information.

Processing all 1,000 features requires more:

- Computation
- Memory
- Processing time

3. Analysis

A large feature vector can contain redundant information. Features that provide similar information do not all need to be retained.

This creates a **high-dimensional feature space**, making classification and matching more expensive.

4. Proposed Solution

Apply **Principal Component Analysis (PCA)** to transform the original features into a smaller number of principal components.

1,000 Original Features → PCA → 100 Principal Components

PCA selects components that capture the largest amount of variance in the original dataset.

5. Expected Result

The system processes fewer features while retaining most of the important information.

Benefits include:

- Reduced computational cost
- Reduced memory usage
- Removal of redundant information
- Faster feature processing

- More efficient classification

6. Conclusion

PCA effectively reduces the dimensionality of a large feature set while preserving important information. Therefore, it improves the computational efficiency of computer vision systems such as face recognition.